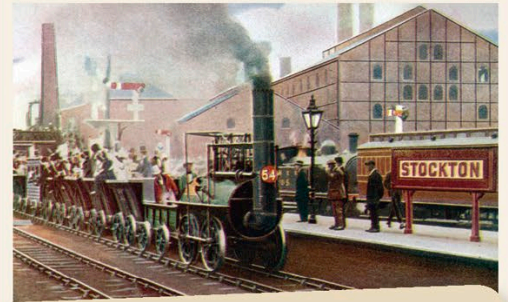


Railroads

While early carriages relied on human power or animals (horses or donkeys) to move them along tracks, it was the invention of the steam locomotive that spurred the development of railroads. Early steam engines (see pp.52–53) were fixed in place and mainly ran pumps and machines in factories, but they were too bulky to use in locomotives. The breakthrough for railroads came in about 1800 with the development of small, powerful, high-pressure steam engines.



THE FIRST PUBLIC RAILROAD

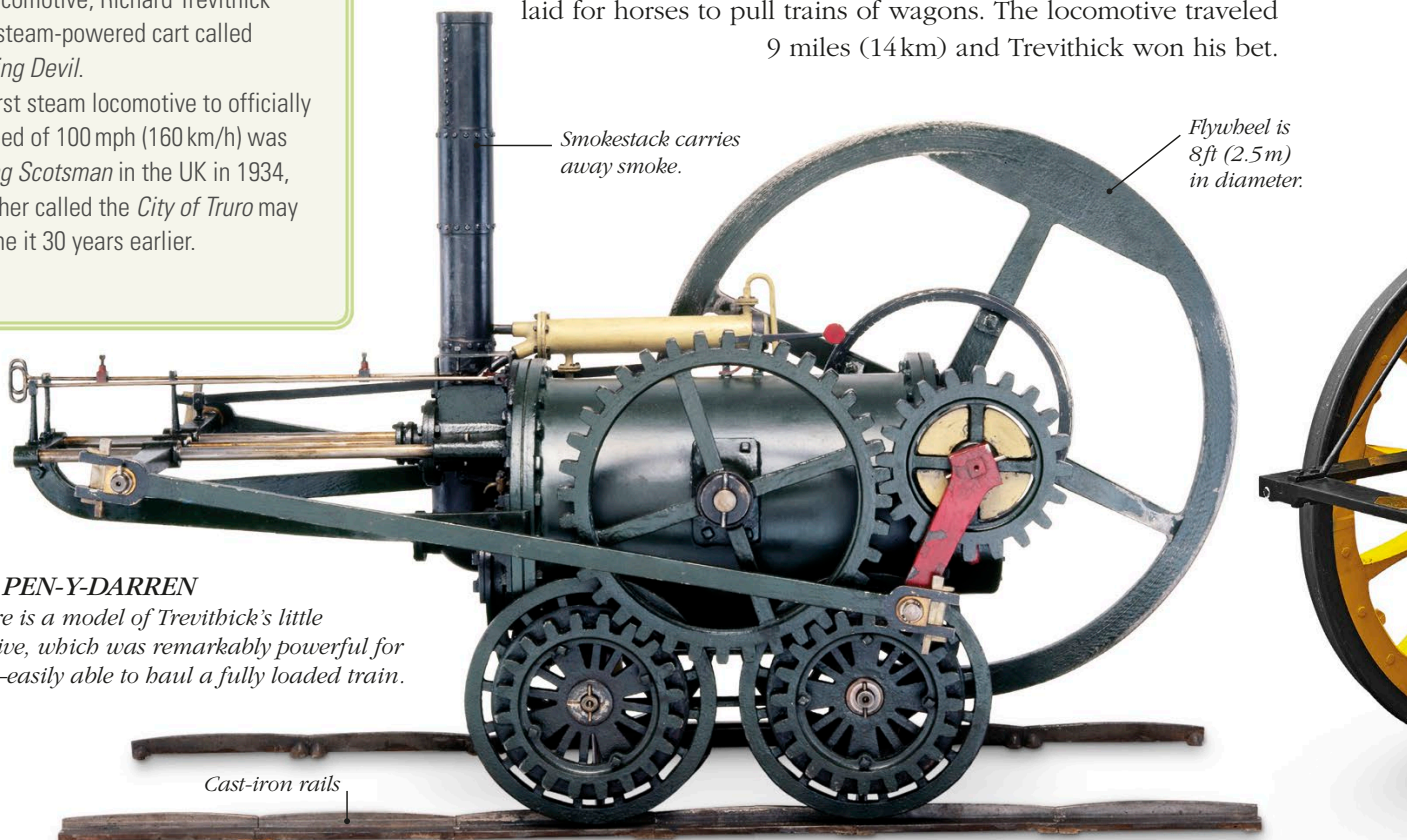
In 1825, Stockton and Darlington, UK, became the first railroad line to open to the public. It was built to carry coal, but on the day it opened, people jumped into the open railcars and rode all the way. The 36 cars were pulled by a locomotive called *Locomotion No. 1*, and carried coal, flour, workmen, and passengers. Seen above is a 1925 reenactment of the event.

FAST FACTS

- The first high-pressure steam engine was built by the American inventor Oliver Evans in the 1790s.
- Three years before he built the first steam locomotive, Richard Trevithick made a steam-powered cart called the *Puffing Devil*.
- The first steam locomotive to officially hit a speed of 100 mph (160 km/h) was the *Flying Scotsman* in the UK in 1934, but another called the *City of Truro* may have done it 30 years earlier.

STEAMING AHEAD

In 1804, the British engineer Richard Trevithick invented the world's first steam locomotive, *Pen-y-Darren*. Trevithick fitted it with his own high-pressure steam engine, and to prove that it worked, he bet it could haul 11 tons of coal along rail tracks laid for horses to pull trains of wagons. The locomotive traveled 9 miles (14 km) and Trevithick won his bet.



► THE PEN-Y-DARREN

Seen here is a model of Trevithick's little locomotive, which was remarkably powerful for its size—easily able to haul a fully loaded train.